

RWKV-TS+: Adaptive Long-range Prompt for Multiple Time Series Tasks

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Abstract—Capturing long-range temporal dependencies plays a critical role in enhancing the prediction performance of downstream tasks. Existing approaches commonly employ classical attention mechanisms to learn interactive relationships, while RWKV-TS further applies linear attention mechanisms to achieve lower computational costs. However, these algorithms model interaction information only within individual batches, falling far short of the theoretical perceptual length limit and thereby hindering the integration of truly long-range historical information. Inspired by prompt learning, we propose the RWKV-TS+ framework, which constructs cross-batch feature interaction patterns without introducing any additional parameters, achieving integration of longer-range information. The modeled sequence length is extended by over two orders of magnitude compared to existing advances. RWKV-TS+ introduces three core modules: the Patch Features Collector integrates intra-batch sequential features; the Time Features Aggregator facilitates inter-batch information interaction; and the Long Prompt Operator projects historical batch sequence information into prompt vectors and injects them into next batch, enabling awareness of longer-range dependencies. Comprehensive experiments on four downstream tasks (long-term forecasting, time series classification, anomaly detection, and imputation) demonstrate that RWKV-TS+ achieves significant performance improvements across multiple benchmarks, with time series classification accuracy improved by up to 28.3%. RWKV-TS+ exhibits unique advantages in modeling ultra-long-range dependencies with adding none additional parameters.

Index Terms—Multivariate time series, Long-range Temporal modeling, Prompt Learning, RNNs-based modeling

I. INTRODUCTION

Time series analysis (TSA) is a fundamental task with immense practical value, widely employed in diverse domains such as weather forecasting [1], [2], industrial anomaly detection [3], [4], traffic flow estimation [5], [6], and financial investment [7]. Unlike other sequential data such as natural language or video, time series are characterized by a continuous recording of scalar values at each time point, where individual observations often lack sufficient standalone semantic meaning.

The primary focus of TSA lies in extracting and understanding meaningful temporal variations, such as trends and seasonal fluctuations. However, real-world time series frequently exhibit intricate and overlapping temporal patterns, making variation modeling particularly challenging. Recently,

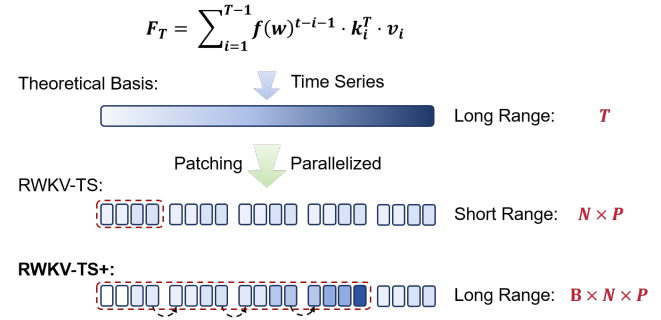


Fig. 1. Visualization of difference comparison of sequence length capturing temporal information between Theoretical Basis, RWKV-TS, and RWKV-TS+ (ours).

deep learning models have made significant progress in time series representation and analysis. Representative methods capture temporal variations and semantics through specialized architectures, spanning several foundational backbones.

Multilayer Perceptrons (MLPs), as classic neural networks, encode temporal dependencies via the fixed weights of MLP layers. MLP-based models have demonstrated competitive performance in TSA [8]. For instance, DLinear and its variants decompose raw sequences into trend and remainder components, employing simple linear layers to learn temporal features. Another category of methods utilizes Convolutional Neural Networks (CNNs) along the temporal dimension (TCNs) to extract variation information [9], [10]. Notably, TimesNet [11] transforms 1D time series into sets of 2D tensors based on multiple periodicities, leveraging 2D kernels to model complex multi-periodic dependencies. Inspired by advancements in Natural Language Processing (NLP), Transformer-based methods have been extensively developed for time series tasks [12], [13]. Among these, Autoformer [14] incorporates decomposition and auto-correlation concepts from classical TSA, while PatchTST [15] segments time series into smaller patches to serve as input tokens. However, their quadratic computational complexity $\mathcal{O}(L^2)$ and limitations in capturing ultra-long-term dependencies hinder their ability to model non-linear temporal dynamics in extremely long sequences.

Traditional time series modeling frequently employs Recurrent Neural Networks (RNNs) [16] and their prominent extensions, including Elman RNN, Long Short-Term Memory (LSTM) [17], and Gated Recurrent Units (GRU). While these

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models are designed to learn temporal patterns from sequential observations, they frequently fail to capture long-term dependencies owing to the vanishing and exploding gradient phenomena. Moreover, the iterative nature of the RNN paradigm precludes massive parallelization during training, imposing a bottleneck on computational throughput. To overcome these constraints, the RWKV-TS architecture has recently emerged. This model successfully maintains the capability to capture long-range temporal dependencies while maintaining the efficiency of RNNs, characterized by its $\mathcal{O}(L)$ time complexity during inference and parallelizable training efficiency.

Exploring the architecture and intrinsic mechanism of RWKV-TS, we find that a linear attention mechanism is employed to model long-range temporal interactions with lower computational cost. Furthermore, the model achieves recursive propagation of information under parallel training conditions. Theoretically, as illustrated in Figure 1, WKV operator constructs a linear attention mechanism augmented with time decay vector, enabling dependency aggregation over sequences of length T . However, in practical implementation (seeing open-source code), the batching and patching strategies result in the independence between each batch. The real size of look-backed window is constrained to $N \times P$ in a single batch, where N denotes the number of patches per batch and P represents the temporal length of each patch. Notably, $N \times P \ll T$, which impedes the model's ability to capture effectively long-range dependencies. Moreover, extending this analysis, we argue that batch-based progression methods [15], [14], [13] in general face analogous limitations, restricting the integration of longer historical information.

To address with these intractable inherent challenges, we propose a novel and efficient framework, termed RWKV-TS+, which enables cross-batch information propagation compared to the vanilla RWKV-TS. Our approach effectively addresses the cold-start problem of the initial patch within a single batch while facilitating longer-range dynamic information aggregation. The model achieves a temporal semantic aggregation length of $B \times N \times P$ (where B denotes the batch size), representing a B -fold improvement over the original RWKV-TS, and up to $T/(B \times P)$ times enhancement under maximum configuration. Furthermore, RWKV-TS+ maintains exactly the same number of parameters as the baseline model, introducing no additional learnable parameters and thereby preserving model compactness.

Specifically, RWKV-TS+ incorporates three key novel modules: the Patch Features Collector introduces boost vectors to aggregate intra-batch temporal information and extract salient features within each batch; the Time Features Aggregator constructs inter-batch WKV operations to capture temporal dependencies across multiple batches; the Long Prompt Operator integrates prompt vectors fused from multiple historical batches into the “next batch”, facilitating cross-batch transmission of temporal information and enabling patches to perceive longer-range context. These three modules operate collaboratively in a progressive manner, bridging and propagating temporal information across otherwise isolated batches. We conduct extensive experiments on four downstream tasks: long-term forecasting, time series classification,

anomaly detection, and imputation. Comprehensive results on multiple benchmark datasets demonstrate that RWKV-TS+ achieves competitive performance compared to several state-of-the-art (SOTA) baselines. Notably, our model improves time series classification accuracy by up to 28.3% over the original baseline. These results underscore the unique strengths of RWKV-TS+ in modeling ultra-long-term temporal dependencies and capturing critical temporal semantics. To summarize, our main contributions are four-fold:

- We propose the RWKV-TS+ model, which achieves cross-batch information capture, learns longer temporal features, and approaches the theoretical maximum integrable sequence length.
- We introduce an inter-batch temporal feature aggregator and a long-range promptor, which collectively gather and integrate temporal features across multiple batches while adaptively incorporating historical prompt to feature learning within each batch.
- The novel operations are free of additional parameters, not only enabling parameter reuse, but also enhancing the model's capacity to capture multi-scale temporal semantics.
- Extensive experiments on multiple downstream tasks and comparisons with various SOTA benchmarks demonstrate the superiority and competitiveness of RWKV-TS+.

II. RELATED WORK

A. Time Series Representation Learning and Analysis

Time series representation learning with Artificial Intelligence has been a prominent research area for many years. Historical data on multi-domain are frequently used to train models for various applications. Some of the earlier techniques in time series forecasting relied on statistics and mathematical models like SARIMA [18], and TBATs [19]. These used moving average and seasonal cycles to capture the patterns for future prediction. With the advent of machine learning, new approaches using Linear Regression and ensemble of decision trees [20] focuses on correcting the errors via quadratic mean loss function. However, these methods still require manual feature engineering and model designs, which is weaker than power of Deep Neural Network (DNN) in terms of efficiency and accuracy.

In recent years, transformer-based architectures is one of the most successful methods for Natural Language Processing (NLP) and have become the basis of Large Language Model (LLM). Their success in NLP has also catalyzed algorithmic advancements in other domains, such as image processing and speech recognition. Besides, applying Transformer to model and analyze time series has emerged as a prominent research direction. Among these, LogTrans avoids a point-wise dot product between the key and query, but its value is still based on a single time step. Informer [12] proposes a ProbSparse self-attention mechanism combined with distillation operations to efficiently extract salient information. Autoformer [14] integrates series decomposition and auto-correlation mechanisms inspired by classical time series analysis, but it is a handcrafted design which doesn't include

all the semantic information with a patch. FEDformer [13] employs a Fourier-enhanced structure to achieve linear computational complexity. More recently, PatchTST [15] segments time series into patches, which are treated as input tokens for transformer-based modeling. Another state-of-the-art approach, iTransformer [21], inverts the embedding process by treating each variate independently. Specifically, it embeds the time points of individual series into variate tokens, which are then processed through self-attention to capture inter-variate dependencies.

Recurrent neural networks (RNNs) based methods summarize the past information compactly in internal memory states and recursively update themselves for forecasting [22]. Traditional RNNs conduct inference with lower efficiency due to non-parallelization training. Meanwhile, they encounter a noticeable bottleneck when the sequence length exceeds 100, where the token loss cannot be further reduced [23]. C2FAR [24] is coarse-to-fine autoregressive networks based on RNN to simultaneously handle discrete and continuous series of arbitrary scale and distribution shape. However, in Long-Term Sequence Forecasting (LTSF) tasks characterized by extended historical windows and prediction horizons, RNNs suffer from limitations in capturing long-term dependencies, which has led to their gradual abandonment [12], [13]. The emergence of RWKV-TS [25] aims to challenge and change this situation, attempting to address these limitations. It integrated the Patching operation and basic RWKV [25] blocks including time-mixing and channel-mixing sub-block to enhance information capturing ability with longer sequence. Moreover, RWKV-TS can be characterized by $\mathcal{O}(L)$ time complexity and memory usage. The Matrix Long Short-Term Memory (mLSTM) [26] model introduces a matrix memory cell along with a covariance update mechanism for key-value pair storage which significantly increases the model's memory capacity. On the other hand, novel RNN models like RWKV [25] and RNN variants such as Mamba [27], [28] and Retentive Network [29] have also exhibited superior efficiency in the domain of LLM. To further improve the RWKV-TS's power of addressing time series tasks with long-term dependencies, we introduce the prompt learning strategy into time-mixing sub-block, termed RWKV-TS+. This extends the sequence length encoded by RWKV-TS by over two orders of magnitude.

B. Adaptive Prompt in Time Series Analysis

The Prompt is an input template or format deliberately engineered by researchers to facilitate downstream tasks. Its primary function is to assist Pre-trained Language Models (PLMs) in recalling and activating the knowledge acquired during the pre-training phase [30]. Prompt templates are primarily constructed through two modes: manual design and automated learning. Handcrafted prompt templates are relatively intuitive but heavily rely on the extensive expertise of specialists. The automated learning paradigm can be further categorized into discrete and continuous approaches. The discrete approach primarily involves statistical learning from a large-scale corpus, such as Wikipedia, to capture

dependency path prompts between input x and output y [31], [32]. In contrast, the continuous approach is not constrained by human-interpretable natural language but rather utilizes model-understandable embedding prompts that align with task-specific training preferences [33], [34], [35].

In the field of time series analysis, the exploration of prompt design—leveraging inductive biases for temporal reasoning and knowledge sharing—is still in its nascent stages [33]. PromptCast [36] pioneered the integration of prompts with Large Language Models (LLMs) for TSF by transforming numerical inputs into natural language templates, enabling sentence-to-sentence forecasting. Similarly, GPT4MTS [37] employs a prompt-based framework to guide LLMs in processing numerical time series data. LLM-Time [33] directly textualizes temporal data as hard prompts; however, this approach often incurs feature redundancy due to the high overlap between textual descriptions and raw numerical sequences.

Unlike handcrafted textual prompts, TEMPO [38] decomposes complex temporal interactions into trend, seasonal, and residual components, further utilizing a selection of top-K prompts to facilitate distribution adaptation across diverse time series. Such soft prompts allow for the efficient fine-tuning of Generative Pre-trained Transformers (GPT). S2IP-LLM [39] aligns pre-trained semantic spaces with time series embedding spaces, performing forecasting based on prompts learned from the joint space. Recently, TSF-Prompt [40] integrated multi-prompt information and cross-modal semantic alignment to enhance LLM-based forecasting. Despite these advances, existing methods primarily rely on static prompts, which fail to adapt dynamically to rapid temporal variations. In this work, we develop a dynamic prompt construction mechanism that enables adaptive prompting tailored to evolving sequential data. By incorporating historical state features into the prompts, we enhance both the expressive capability and the computational efficiency of the prompt vectors.

III. METHODOLOGY

RWKV-TS+ can be schematically summarized as comprising three core components: Patch Features Collector, Time Features Aggregator and Long Prompt Operator, as follows the figure 2. The Patch Features Collector extracts sequential features from the patch matrix within each input batch, generating densely aggregated summary representations at the intra-batch level. The Time Features Aggregator is primarily responsible for modeling dependencies across summary representations from multiple batches, thereby constructing integrated long-range temporal cues that leverage cross-batch information. Long Prompt Operator integrate the prompt into the next batch series, realizing the temporal features passing across batches. The figure 3 exhibit the that the prompt embeddings collect the historical information and serve to the current series.

A. Patch Features Collector

Given the trained input sequence data, $X \in \mathbb{R}^{B \times T \times C}$ where B, T, C are the batch size, patch size, and sequence length, respectively. the sequence of patches $X(b, t)$ is firstly encoded to the input tokens $x_{b,t} \in \mathbb{R}^D$ through a learnable projection

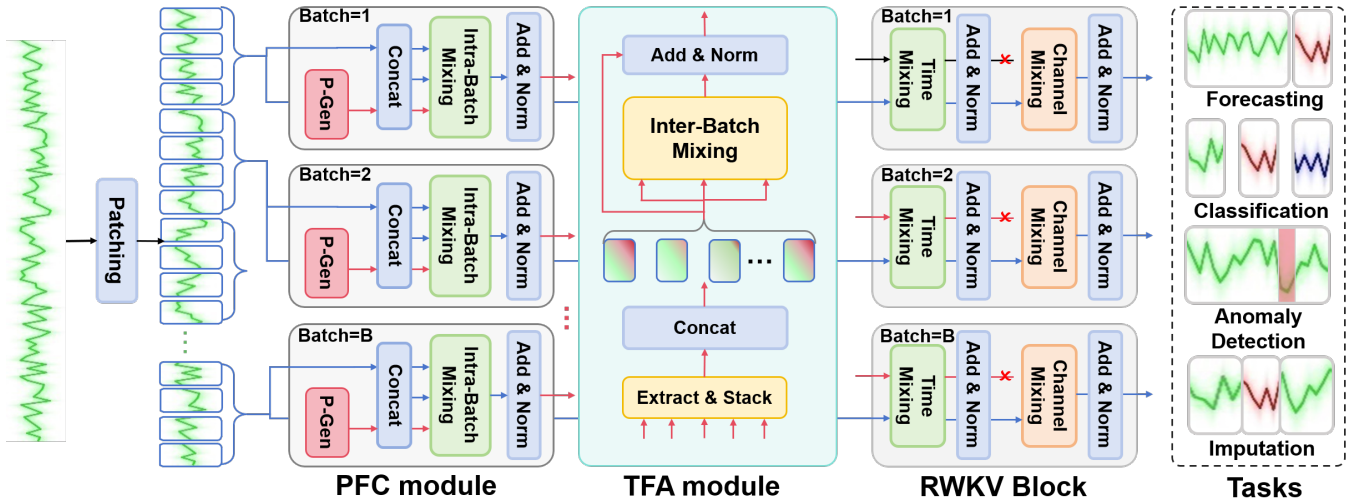


Fig. 2. The overall architecture of RWKV-TS+. It can be summarized as comprising two core components: the Patch Features Collector (**PFC module**) to extract the information from each batch and the Time Features Aggregator (**TFA module**) to adaptively integrate long-term representation. P-Gen stands for Prompt Generator to initialize the prompt embedding vectors. RWKV-TS+ can serve various temporal downstream tasks beyond forecasting.

matrix, where D is embedding dimension of the whole backbone. We predefine the prompt embedding $X^{pt} \in \mathbb{R}^{B \times 1 \times D}$ for each batch. To realize the sequence interaction cross batches, the features of multiple patches in a single batch b firstly are integrated and extracted via the time mixing sub-block,

$$M_b^{pt} = W_M \cdot (\mu_M \odot x_b^{pt} + (1 - \mu_M) \odot x_{b,T}), \quad (1)$$

where $W_M, M \in \{g, r, k, v\}$ denotes matrix multiplications cells, M_b^{pt} are the embedding set of gate, receptance, key, and value projection vectors. $\mu_M, M \in \{g, r, k, v\}$ are the proportional factors. After the multi-head WKV operator and output gating, we can obtain intra-batch sequence feature h_b to represent multi-patch information in the b -th batch. stacking B intra-batch features, we get the embedding matrix $H = \{h_1, h_2, \dots, h_b, \dots, h_B\} \in \mathbb{R}^{B \times D}$ during one trained cycle.

B. Time Features Aggregator

Time features aggregator aim to realize the interaction of sequence features h_b between batches. we expect that batch features are progressively backward integrated through a time-dependent decay factor, which is satisfied with the temporal causality property. Apparently, the aggregation process is the operation of time mixing sub-block. Thus we straightforward reemploy the time mixing module of corresponding block layer, as follows,

$$g_b = W_g \cdot (\mu_g \odot h_b + (1 - \mu_g) \odot h_{b-1}), \quad (2)$$

$$r_b = W_r \cdot (\mu_r \odot h_b + (1 - \mu_r) \odot h_{b-1}), \quad (3)$$

$$k_b = W_k \cdot (\mu_k \odot h_b + (1 - \mu_k) \odot h_{b-1}), \quad (4)$$

$$v_b = W_v \cdot (\mu_v \odot h_b + (1 - \mu_v) \odot h_{b-1}). \quad (5)$$

The self-attention can help the batch feature h_b locate to the crucial historical information from the prior $b - 1$ batch features. We still employ the linear-time self-attention operator to calculate the dependence weights wkv_b of different

batches. In fact, the current token information and recent historical information always present the core influence. We extra add the channel-wise temporal decay vector $\text{diag}(e^w)$ for prior batch features and learnable reward vector $\text{diag}(u)$ for current batch feature. Combining different weight operators to compute the wkv_b as follows:

$$wkv_b = \text{diag}(u) \cdot k_b^T \cdot v_b + \sum_{i=1}^{b-1} \text{diag}(e^w)^{b-1-i} \cdot k_i^T \cdot v_i, \quad (6)$$

where $w = \exp(-w)$ to limit the all values within the range (0,1), ensuring that $\text{diag}(e^w)$ denotes a contraction matrix. Similarly, we introduce gating vector g_b and receptance vector r_b to compute the final WKV -based aggregate output. The output vector c_b is formally described by the following equation:

$$c_b = \text{SiLU}(g_b) \odot (\text{LayerNorm}(r_b \cdot wkv_b))W_o, \quad (7)$$

where SiLU is a non-linear projection function used in vanilla RWKV-TS. The residual connection is adopted to integrate the original intra-batch features h_b and inter-batch features c_b as follows:

$$p_b = \text{LayerNorm}(h_b + c_b). \quad (8)$$

C. Long Prompt Operator

Constructing the historical prompt embeddings mainly mask up for the lack of information perception in the current batch. As shown in Fig.3, after obtaining the inter-batch information embedding p_{b-1} , we concatenate the p_{b-1} and the original trained input X_b , $[p_{b-1}, X_b] \in \mathbb{R}^{(T+1) \times D}$ to introduce the historical information for the b -th batch data. The time mixing sub-blocks are reused to get the new patch representations with the prior information. Specifically, The formula of four core vectors $M_{b,1}$ of mixing long prompt when $t = 1$ in the b -th batch are given by:

$$M_{b,2} = W_M \cdot (\mu_M \odot x_{b,1} + (1 - \mu_M) \odot p_{b-1}), \quad (9)$$

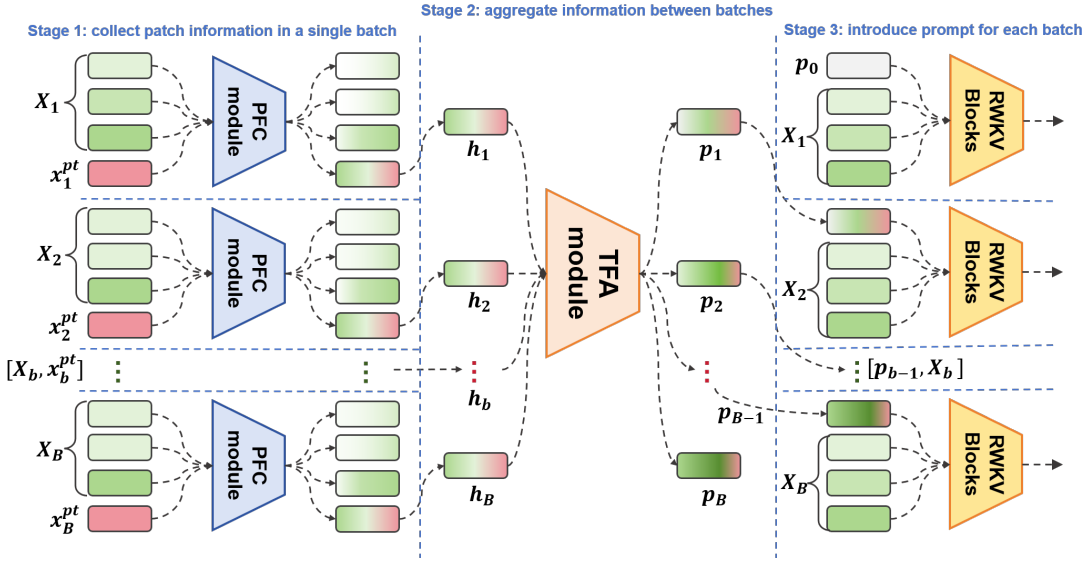


Fig. 3. The process of generating representations with long-range temporal information and performing cross-batch prompting operations. x_b^{pt} denotes the initialized embedding serving as a container for historical information and p_0 is a zero-padded vector.

however, when $t > 1$, we still adopt the token shift, a simple and intuitive way of "time mixing", to learn the core vectors:

$$M_{b,t} = W_M \cdot (\mu_M \odot x_{b,t} + (1 - \mu_M) \odot x_{b,t-1}). \quad (10)$$

In Practice, the token shift operator are preformed by parallel computation, the Conducting the multi-head WKV operator and output gating of the rest of time mixing sub-block, we can obtain the time mixing patch embeddings $O_b \in \mathbb{R}^{(T+1) \times D}$ in the b -th batch. However, the $o_{b,1}$ is only a unmeaning vector introducing the long prompt. In order to minimize the effect and keep the original the mode of channel mixing sub-block in RWKV-TS, we remove the $o_{b,1}$ from O_b embeddings, $O'_b = O_b / o_{b,1} \in \mathbb{R}^{T \times D}$. Note that while the Long Prompt Operator is designed for sequential continuity, its application in classification (each sequence is treated as independent instances) is adapted to capture global distributional invariants.

The entire aforementioned process is reused into multiple RWKV blocks. Following the RWKV-TS architecture, the decoder still employs a flatten layer with linear head to predict the output y_i . The mean squared error (MSE) loss remains adopted as the loss function.

D. Heuristic Analysis of Information Aggregation

Assuming the information of each patch feature x_t is defined as $I(t)$. When RWKV-TS is trained via the parallelized category, the most information that the patch feature can aggregate is $I(P)_{intra} = \sum_{t=1}^T I(n)$, where N is the number of patch in each batch. However, RWKV-TS++ introduces X_{prompt}^b , a information memory vector, to encode the information of single batch and conduct the high aggregation between multiple batches. In this work, the WKV operator can compute the aggregation weights of multiple patches cross batches. for ensured the patch $x_{b,t}$ in the b -th batch, the captured historical information $I(b, t)$ is view as the function of $wkv'_{b,t}$.

Conceptual Formulation 1 (Equivalence of WKV with long prompt). Supposing the time decay factor is the function $f_d(\cdot)$ of e^w and the number of patches. For given the patch $x_{b,t}$, combining the long prompt, the $wkv'_{b,t}$ can be writing as:

$$wkv'_{b,t} \sim f(u) \cdot k_{b,t}^T \cdot v_{b,t} + \sum_{j=1}^{bT-T+t-1} f_d(e^w, bT-T+t-1, j) \cdot k_j^T \cdot v_j. \quad (11)$$

Based on the above analysis, the most information that the patch feature can obtain is $I(P)_{inter} = \sum_{b=1}^B F_b(\sum_{t=1}^T I(n))$. theoretically, RWKV-TS+ extends the potential temporal receptive field by up to B times compared to the vanilla RWKV-TS, providing a significantly larger capacity for capturing long-range dependencies. The derivation process for the approximation of this conceptual formula is reported in Supplementary Material.

E. Model Complexity

According to the model framework of RWKV-TS reported in the literature and code, the parameters mainly consist of embedded layer, multiple RWKV blocks, and output layers. The total number of trainable parameters, N_{params} , are computed as:

$$N_{params} = N(embed) + \sum_{blocks} (\sum_{heads} (N(wkv))) + N(clf). \quad (12)$$

Similarly, the parameters of RWKV-TS+ adds extra two modules, prompt embedding layer and prompt time mixing sub-blocks.

$$N_{params} = N(embed) + \sum_{blocks} (\sum_{heads} (N(wkv)) + N(pt_embed) + \sum_{heads} (N(pt_TM))) + N(clf). \quad (13)$$

The predefined prompt embeddings could be not updated, the parameters of X_{prompt} are zero, $N_{params}(pt_embed) = 0$. And the Time Mixing sub-block for prompt embeddings are shared parameters with the time mixing sub-block in patch RWKV block for patch embeddings, the extra trainable parameters, $N(pt_TM)$, are zero, $N_{params}(\sum^{head}(N(pt_TM))) = 0$. We finally can infer that the parameters of RWKV-TS+ are equal to the RWKV-TS ones, $N_{params}^{RWKV-TS+} = N_{params}^{RWKV-TS}$.

IV. IMPLEMENTATION DETAILS

To evaluate the reliability and generalizability of the proposed RWKV-TS+, this study selects four downstream tasks that require modeling of long-range temporal dependencies, including long-term forecasting, classification, anomaly detection, and imputation. For each downstream task, multiple representative benchmarks are adopted for comprehensive experimental validation. Several representative approaches are selected and compared RWKV-TS+ over the performance and efficiency. Detailed parameter configurations across different tasks and datasets are provided in the subsection IV-C and supplementary materials.

A. Baseline Models

To accommodate diverse downstream tasks, we selected eight universal approaches for which comprehensive experimental studies were conducted on time series. Their experimental results across different tasks are referenced from [41]. Based on their core frameworks, these advances can be categorized into:

- CNNs-based methods: TimesNet [11];
- MLPs-based methods: DLinear [8] and LightTS [42];
- Transformers-based methods: Informer[12], PatchTST[15], Autoformer[43], Stationary[44], FEDformer[13], and ETSformer[45];
- RNNs-based methods: RWKV-TS [41].

Additionally, we also selected task-specific approaches to compare with RWKV-TS+. For anomaly detection tasks, we added the Anomaly Transformer [3] as baseline. For the classification task, we added several models for the two categories: Rocket and XGBoost [20] for classical methods; LSTM, LSTNet [46], and LSSL [47] for RNNs-based methods. To validate the efficiency of the designed prompt in RWKV-TS+, we compare specifically with several temporal models using prompt learning: TSF-Prompt [40], Time-LLM [48], TEMPO [38].

B. Datasets & Metrics

The table I is a summary of benchmarks, and reports the used benchmarks, metrics and series length for four tasks. The series lengths of forecasting and imputation represent the predicted length. The subsets in ETT datasets are ETTh1, ETTh2, ETTm1, ETTm2. The ten subsets of UEA datasets are EthanolConcentration (EC), FaceDetection (FaceD), Handwriting (Handw),

Heartbeat, JapancesVowels (JV), PEMS-SF, SelfRegulation-SCP1 (SR-SCP1), SelfRegulationSCP2 (SR-SCP2), SpokenArabicDigits (SAD), UWaveGestureLibrary (UWGL). Different metrics match different downstream tasks. The mean squared error (MSE) and mean absolute error (MAE) are for Long-term forecasting and Imputation task. The classification task is evaluated via accuracy metric. The precision, recall, and F1-score metrics is selected for Anomaly Detection task.

C. RWKV-TS+ Settings

Building upon the baseline RWKV-TS framework, we maintain the original dataset partitioning. All the experiments are implemented in PyTorch and conducted on a single NVIDIA 3090 24GB GPU. During the data loading phase, to preserve contextual continuity within sequence information across batches, the shuffle parameter in the data loader function is disabled, ie. `torch.utils.data.DataLoader(data, shuffle = False)`. Furthermore, to achieve optimal performance, we fine-tune several parameters in RWKV-TS for different downstream tasks. Specifically, for long-term forecasting tasks, we adjust the batch size, input sequence length (seq_len), and label sequence length (label_len); for classification tasks, we optimize the learning rate (lr), patch size, and training epochs; for anomaly detection tasks, only the batch size is fine-tuned; and for time series imputation tasks, both input sequence length and batch size are adaptively configured. Given that the batch size critically determines the model's look-back window length and the extent of historical information integration, we provide a comprehensive ablation study in the experimental section, seeing subsection V-E1. Detailed parameter configurations for each dataset are reported in the supplementary materials. The code can be found with <https://github.com/xinhuaxi/RWKV-TS-Plus>.

V. EXPERIMENT RESULTS AND ANALYSIS

A. Long-term Forecasting

Long-term time series forecasting aims to utilize historical information to predict data variations at multiple future time points, evaluating a model's capability to extract and model temporal dynamics. In our work, we selected eight widely-used benchmarks to assess the performance of RWKV-TS+, while simultaneously comparing it with 10 SOTA baselines. For each dataset, the optimal MSE and MAE metrics across different prediction lengths are reported in the supplementary materials, with the mean prediction discrepancy metrics recorded in Tab. II. We observe that RWKV-TS+ achieves competitive performance across multiple datasets. PatchTST still achieves the best performance across multiple benchmarks. The average MSE and MAE over these benchmarks are 0.446 and 0.386, respectively, representing a 14.4% reduction in MSE and a 5.9% reduction in MAE compared to RWKV-TS+. However, compared to RWKV-TS, our model exhibits a performance improvement, particularly with a 10.5% error reduction on MSE for the ETTm2 dataset.

We further investigate the advantages of PatchTST and find that it employs the original Transformer architecture, which enables it to learn rich contextual information and accurately

TABLE I
SUMMARY OF EXPERIMENT BENCHMARKS

Tasks	Benchmarks	Metrics	Series Length
Long-term Forecasting	ETT(4 subsets), Electricity, Traffic, Weather, ILI	MSE, MAE	96~720 (ILL: 24~60)
Classification	UEA(10 subsets)	Accuracy	29~1751
Anomaly Detection	SMD, MSL, SMAP, SWaT, PSM	Precision, Recall, F1-Score	100
Imputation	ETT(4 subsets), Electricity, Weather	MSE, MAE	96

TABLE II
COMPARISON RESULTS FOR **LONG-TERM FORECASTING** TASKS WITH 10 BASELINES. ALL THE RESULTS ARE AVERAGED FROM 4 DIFFERENT PREDICTION LENGTHS, THAT IS {24, 36, 48, 60} FOR ILI AND {96, 192, 336, 720} FOR THE OTHERS. THE BEST MSE AND MAE ARE BOLD AND RED AND THE 2ND BEST ONES VIA BOLD AND BLUE.

Methods	ETTh1		ETTh2		ETTm1		ETTm2		Weather		ILI		ECL		Traffic		Average	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Informer	1.040	0.795	4.431	1.729	0.961	0.734	1.410	0.810	0.634	0.548	5.137	1.544	0.311	0.397	0.764	0.416	1.836	0.871
Autoformer	0.496	0.487	0.450	0.459	0.588	0.517	0.327	0.371	0.338	0.382	3.006	1.161	0.227	0.338	0.628	0.379	0.757	0.511
Stationary	0.570	0.537	0.526	0.516	0.481	0.456	0.306	0.347	0.288	0.314	2.077	0.914	0.193	0.296	0.624	0.340	0.633	0.465
PatchTST	0.413	0.430	0.330	0.379	0.351	0.387	0.255	0.315	0.225	0.264	1.443	0.798	0.161	0.253	0.390	0.264	0.446	0.386
FEDformer	0.440	0.460	0.437	0.449	0.448	0.452	0.305	0.349	0.309	0.360	2.847	1.144	0.214	0.327	0.610	0.376	0.701	0.489
DLinear	0.423	0.437	0.431	0.447	0.357	0.378	0.267	0.334	0.249	0.300	2.169	1.041	0.166	0.263	0.434	0.295	0.562	0.436
LightTS	0.491	0.479	0.602	0.543	0.435	0.437	0.409	0.436	0.261	0.312	7.382	2.003	0.229	0.329	0.622	0.392	1.303	0.616
ETSformer	0.542	0.510	0.439	0.452	0.429	0.425	0.293	0.342	0.271	0.334	2.497	1.004	0.208	0.323	0.621	0.396	0.662	0.473
TimesNet	0.458	0.450	0.414	0.427	0.400	0.406	0.291	0.333	0.259	0.287	2.139	0.931	0.192	0.295	0.620	0.336	0.596	0.433
RWKV-TS	0.433	0.445	0.375	0.412	0.376	0.401	0.287	0.338	0.231	0.266	1.910	0.925	0.159	0.253	0.398	0.276	0.521	0.414
RWKV-TS+	0.444	0.446	0.352	0.392	0.355	0.388	0.257	0.317	0.249	0.290	1.961	0.931	0.160	0.252	0.392	0.267	0.521	0.410
↓ <i>RWKV-TS</i>							↓ 10.5%										-	↓ 0.9%

capture temporal variation trends during the train process. This non-causal design lacks future masking, which permits "look-ahead" behavior where early patches aggregate information from subsequent patches within the same window. While this boosts static benchmark performance, it risks capturing spurious correlations that violate strict temporal causality. In contrast, the RNN-based RWKV-TS+ exclusively integrates historical sequence information and dynamically leverages historical feature experience for long-term forecasting. Moreover, compared to RWKV-TS, our model further can utilize longer temporal information. Figure 4 visualizes the cross-patch attention weight heatmaps of PatchTST and RWKV-TS+ for an arbitrary batch. Clearly, non-zero attention weights exist between $patch_1$ and $patch_{64}$ in Fig.4(a), violating temporal causality and indicating information leakage. In contrast, figure 4(b) exhibits no such leakage. Simultaneously, we observe that $patch_1$ in Fig.4(b) (i.e. the prompt embedding) exhibits dynamic and significant attention weights across subsequent patches, serving as a "distilled historical anchor." This proves that our prompt operator successfully provides a meaningful context distilled from preceding batches, effectively extending the model's temporal horizon without compromising causal integrity.

To evaluate the accuracy of long-range forecasting, we

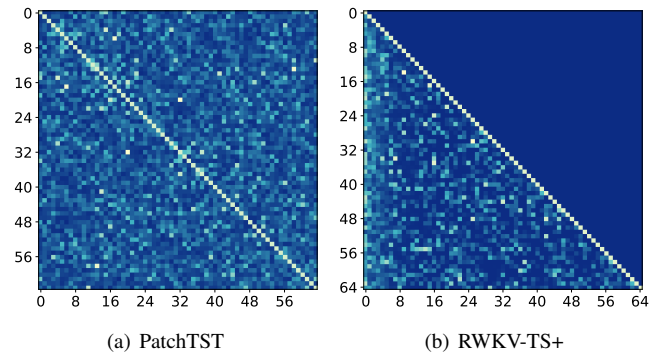


Fig. 4. Heatmap of patches for PatchTST and RWKV-TS+ models in **Long-term Forecasting** task.

meticulously visualized the outcomes of four distinct time-series prediction methods, as presented in Fig.5. RWKV-TS+, alongside two SOTA baselines, PatchTST and RWKV-TS, were selected for comparative assessment of predictive discrepancies. The predictions shown for the first batch primarily compare the precision of the three models when operating without any prior prompts. As observed in Fig. 5(a), the predictions of all three models exhibit significant deviations from the ground truth. Conversely, in Fig. 5(b), RWKV-TS+,

leveraging historical feature prompts, learns partial underlying patterns and consequently captures the trend of the ground truth. However, it still struggles to achieve precise fits within peak and trough regions of the temporal sequence, a challenge shared by the other baselines. In longer-sequence prediction tasks (sequence length $L=720$), the predictive bias gradually accumulates and magnifies over time (refer to Fig. 5(c)).

B. Time Series Classification

To comprehensively assess the representation capacity of RWKV-TS+ for time series analysis, we conducted experiments within the time series classification paradigm. However, in time-series classification tasks, each batch contains a single sequential sample, and these samples are independent with no temporal coherence between them. Introducing enhanced features from the previous batch into the current batch is therefore meaningless. Given this, we refine the Long-range Prompt Operation Module to concatenate prompt features corresponding to the current batch, $[p_b, X_b]$, for time-series classification. This adjustment aims to leverage the intrinsic attribute features of multiple samples to reinforce the identification of discriminative sample characteristics, thereby improving classification accuracy. Table III reports the results of the refined RWKV-TS+ on 10 representative time-series classification datasets, along with the mean accuracy of these datasets. We compare against 15 baselines categorized 5 distinct model types. The results demonstrate that RWKV-TS+ achieves superior performance on multiple datasets. It attains the highest mean classification accuracy across datasets at 74.9%, outperforming the second-best model, TimesNet (73.6%), by 1.7%. Furthermore, compared to the original RWKV-TS, our model achieves a significant 28.3% improvement on the *HandWriting* dataset.

To evaluate the utility of incorporating long-range prompts in time-series classification tasks, we selected binary classification (SelfRegulationSCP2) and multi-class classification (PEMS-SF and UWGL) datasets for visual presentation and analysis, as shown in Fig. 6 and 7, respectively. The baseline model for comparison was RWKV-TS. Figure 6 exhibits the probability distribution of the 0-1 classification for each test sample by the models, where the dashed line represents the decision boundary. It is evident that the majority of samples in Fig. 6(a) reside within the upper-left quadrant, indicating that RWKV-TS classifies nearly all samples as Class 1, thereby failing to capture inter-class discriminative features. However, RWKV-TS+ leverages the long-range prompt to aggregate global structural priors from previously processed samples. This mechanism prevents the model from falling into a local optimum where it fails to distinguish inter-class boundaries (as seen in the "collapsed" distribution of RWKV-TS in Fig. 6). By maintaining a running memory of discriminative features, RWKV-TS+ facilitates a more dispersive and balanced sample distribution across the decision boundary. Consequently, the classification accuracy shows an improvement of 3.8% compared to RWKV-TS.

We extracted the final-layer embeddings from the model and applied the T-SNE algorithm for dimensionality reduction. Figure 7 visualizes the distribution of test set samples,

where the color of each point represents its ground-truth class label. Compared to RWKV-TS, the proposed model induces more distinct separation between clusters representing different classes (see Fig. 7(a) and (b)). Concurrently, samples belonging to the same class exhibit tighter aggregation, as exemplified by the yellow cluster (Class 1) in Fig. 7(c) and (d). These observations suggest that incorporating class-specific prompts enhances representation learning within the model, thereby improving inter-class discriminability.

C. Time Series Anomaly Detection

We conduct comprehensive experiments in five datasets to compare with several SOTA baselines. These baselines' scores are extracted from previous works. The F1-score results for all experiments are presented in Table IV, while precision and recall metrics can be found in the supplementary materials. We observe that RWKV-TS+ achieves more balanced performance, surpassing the current SOTA model TimesNet by an average F1-score across all five datasets, while simultaneously improving upon the baseline RWKV-TS by 1.6%. Notably, the most significant improvement (7.4%) occurs on the MSL dataset. These findings demonstrate our model's balanced and robust capability in learning diverse data patterns, enhancing its versatility as a multi-functional time-series model. Given that this anomaly detection task employs a reconstruction-based training paradigm, incorporating historical priors enables the model to better predict normal temporal trends.

Nevertheless, we acknowledge that RWKV-TS+ does not achieve optimal performance, but is close to the performance of other SOTA baselines. Potential factors contributing to this observation include: (1) Variability in optimal historical window sizes across datasets. For instance, SMAP signals require shorter look-back windows due to their rapid fluctuation period, whereas MSL benefits from longer temporal contexts. This explanation is corroborated by the design principles of frameworks like DLinear. (2) Most baseline methods fail to account for inter-variable dependencies in multivariate time-series anomaly detection, thereby compromising detection accuracy.

D. Imputation

Time series imputation serves as a key downstream benchmark for evaluating a model's capacity to capture temporal dependencies and patterns. We conducted imputation experiments on six widely used real-world datasets (ETTh1, ETTh2, ETTm1, ETTm2, ECL, Weather), which commonly exhibit missing data. Following the setting adopted in TimesNet, various random mask ratios (12.5%, 25%, 37.5%, 50%) of time points were selected to evaluate different proportions of missing data. Table V reports the mean error rates across different missing rates for individual datasets, along with the overall average error across all datasets. A significant observation is that, compared to RWKV-TS, our model achieves a notable reduction in the multi-dataset average imputation error. Specifically, the MSE decreases from 0.120 to 0.114, getting a 5.0% gain. Furthermore, on the ETTm1 dataset, the MSE

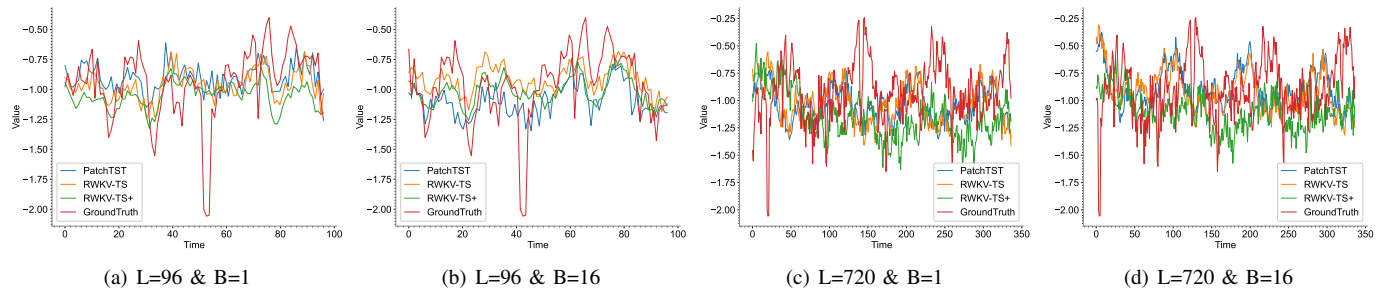


Fig. 5. Visualization of **Long-term Forecasting** results on the test set of ETTm2. L=96/720 represents the predicted length are 96/720, and B=1/16 represents that the 1/16-th batch of test set.

TABLE III

COMPARISON RESULTS OF **ACCURACY** METRIC FOR THE **CLASSIFICATION** TASK WITH 15 MULTI-CATEGORY BASELINES. THE BEST SCORE IS BOLD AND RED AND THE 2ND BEST ONE VIA BOLD AND BLUE. THE ABBREVIATION OF DATASETS ARE THAT EC: ETHANOLCONCENTRATION, FACED: Facedetection, HANDW: HANDWRITING, JV: JAPANCESVOWELS, SR-SCP1: SELFREGULATIONSCP1, SR-SCP2: SELFREGULATIONSCP2, SAD: SPOKENARABICDIGITS, UWGL: UWAVEGESTURELIBRARY.

Methods		EC	FaceD	Handw	Heartbeat	JV	PEMS-SF	SR-SCP1	SR-SCP2	SAD	UWGL	Average
Classical Methods	Rocket	31.1	66.7	24.6	72.7	98.4	86.1	90.8	52.2	100.0	85.9	70.9
	XGBoost	39.9	65.7	25.8	77.1	98.1	86.7	84.0	52.8	100.0	87.8	71.8
MLPs	DLinear	32.6	68.0	27.0	75.1	96.2	75.1	87.3	50.5	81.4	82.1	67.5
	LightTS	29.7	67.5	26.1	75.1	96.2	88.4	89.8	51.1	100.0	80.3	70.4
CNNs	TimesNet	35.7	52.8	32.1	78.0	98.4	89.6	91.8	57.2	99.0	85.3	73.6
	TCN	28.9	-	53.3	75.6	98.9	68.8	84.6	55.6	95.6	88.4	70.3
Transformers	Transformer	32.7	67.3	32.0	76.1	98.7	82.1	92.2	53.9	98.4	85.6	71.9
	Informer	31.6	67.0	32.8	80.5	98.9	81.5	90.1	53.3	100.0	85.6	72.1
	Autoformer	31.6	68.4	36.7	74.6	96.2	82.7	84.0	50.6	100.0	85.9	71.1
	ETSformer	28.1	68.6	32.5	71.2	95.9	86.0	89.6	55.0	100.0	85.0	71.0
RNNs	LSTM	32.3	57.7	15.2	72.2	79.7	39.9	68.9	46.6	31.9	41.2	48.6
	LSTNet	43.7	63.3	15.8	73.2	86.5	98.3	84.6	48.9	69.6	75.9	66.0
	LSSL	45.2	64.7	58.8	75.6	96.2	75.1	90.8	53.3	71.2	94.4	72.5
	RWKV-TS	34.6	67.3	34.6	77.1	98.4	83.8	91.8	57.2	98.9	86.9	73.1
	RWKV-TS+	34.2	68.0	44.4	78.1	98.4	84.4	92.2	59.4	99.3	90.6	74.9
$\uparrow_{RWKV-TS}$		$\uparrow 28.3\%$										$\uparrow 2.4\%$

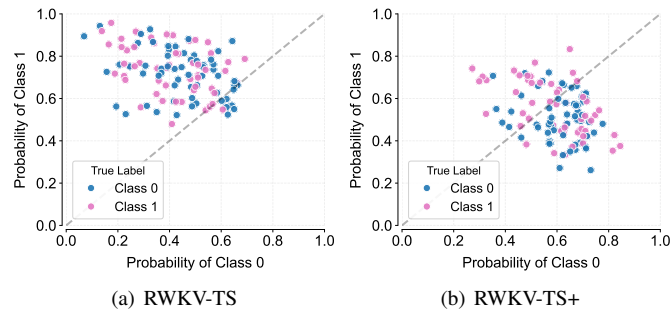


Fig. 6. Scatter of SelfRegulationSCP2 dataset for RWKV-TS and RWKV-TS+ models in **Classification** task. The X-axis represents the probability that the predicted category is 0, and the Y-axis represents the probability that the predicted category is 1.

TS+ remains preforms less effectively than that of SOTA models such as TimesNet and Stationary, though it surpasses MLP-based and certain Transformer-based models.

This outcome suggests two implications. Firstly, incorporating more extensive historical prompt enhances the model's ability to discern sequential patterns, leading to more accurate imputation. Secondly, it highlights an inherent limitation in current RNN-based architectures: their unidirectional nature restricts them to utilizing only information preceding the missing point. In contrast, Transformer-based models are bidirectional, leveraging contextual information both before and after the missing value. This difference can be formally described by the conditional probabilities: $P(z_i | x_1, \dots, x_{i-1})$ for RWKV-TS+, $P(z_i | x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_T)$ for Transformer-based models.

Evidently, the accuracy of temporal imputation exhibits a positive correlation with the volume of contextual information

reduction exceeds 10%. However, the efficiency of RWKV-

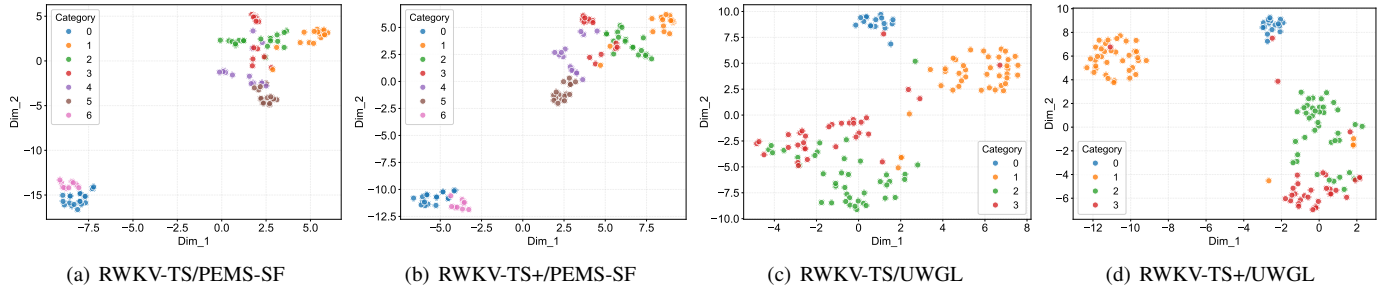


Fig. 7. T-SNE visualization of **Classification** results on the test set of PEMS-SF and UWaveGestureLibrary (UGGL) datasets.

TABLE IV
COMPARISON RESULTS OF **F1-SCORE** METRIC FOR THE **ANOMALY DETECTION** WITH 11 MULTIPLE BASELINES. THE BEST SCORE IS BOLD AND RED AND THE SECOND BEST ONE VIA BOLD AND BLUE.

Methods		SMD	MSL	SMAP	SWaT	PSM	Average
MLPs	DLinear	77.10	84.88	69.26	87.52	93.55	82.46
	LightTS	82.53	78.95	69.21	93.33	97.15	84.23
CNNs	TimesNet	84.61	81.84	69.39	93.02	97.34	85.24
Transformers	PatchTS	84.62	78.70	68.82	85.72	96.08	82.79
	Informer	81.65	84.06	69.92	81.43	77.10	78.83
	Autoformer	85.11	79.05	71.12	92.74	93.29	84.26
	ETSformer	83.13	85.03	69.50	84.91	91.76	82.87
	FEDformer	85.08	78.57	70.76	93.19	97.23	84.97
	Stationary	84.62	77.50	71.09	79.88	97.29	82.08
	Anomaly Transformer	85.49	83.31	71.18	83.10	79.40	80.50
RNNs	RWKV-TS	84.33	77.92	68.71	91.40	97.10	83.89
	RWKV-TS+	84.61	83.71	68.64	92.71	97.18	85.26
$\uparrow_{RWKV-TS}$			$\uparrow 7.4\%$				$\uparrow 1.6\%$

available. We can further infer that when missing data occurs during the early stages of a time series, the imputation performance of RWKV-TS+ underperforms relative to Transformer-based models. Conversely, when missing data occurs at later stages, the performance of the two approaches becomes comparable, i.e., $P_{\text{rnn}}(z_T | x_1, \dots, x_{T-1}) \approx P_{\text{transformer}}(z_T | x_1, \dots, x_{T-1})$. To validate this hypothesis, we present a visual analysis of imputation results on the ETTm1 test set. Figure 8 illustrates the predictions for the first channel of the first batch (sequence length = 64). A clear observation is that PatchTST achieves superior imputation results in the early time steps. However, as the sequence progresses, the imputation quality of RWKV-TS+ gradually surpasses that of PatchTST. This provides evidence for our model's superior capability in learning the intrinsic dynamics of temporal variations.

E. Sensitivity Analysis

1) *Batch Size Setting*: In this work, the batch size governs the effective length of historical sequence information integrated by the RWKV-TS+ model, consequently influencing its performance across various downstream tasks. To evaluate the impact of varying batch sizes on test performance, we selected two representative downstream tasks: long-term forecasting and anomaly detection. Extensive comparative experiments were conducted utilizing four benchmark datasets: ETTm1 and

ETTm2 for long-term forecasting, and SMAP and SWaT for anomaly detection. The corresponding results are documented in Figure 6.

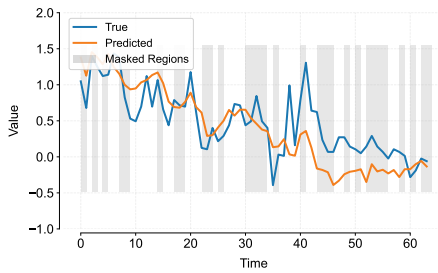
Batch sizes were explored within the range $bs \in \{16, 32, 64, 128, 256, 512, 1024, 2048\}$. On the ETTm1 dataset, the Mean Squared Error (MSE) achieved its minimum value at a batch size of 128. Conversely, the optimal performance on ETTm2 was observed at a batch size of 32. This indicates that ETTm2 exhibits shorter temporal dependencies compared to ETTm1.

For the anomaly detection task, the highest F1-score (68.64%) on the SMAP dataset was attained at a batch size of 512. Notably, the peak Precision (89.83%) on SMAP occurred at $bs = 256$. On the SWaT dataset, the maximum F1-score (92.71%) was only reached when the batch size was increased to 1024. Comparing results across tasks reveals that anomaly detection generally requires modeling over longer historical sequences to capture stable patterns within sequence dynamics. Comprehensive numerical results for all experiments are provided in the supplementary materials.

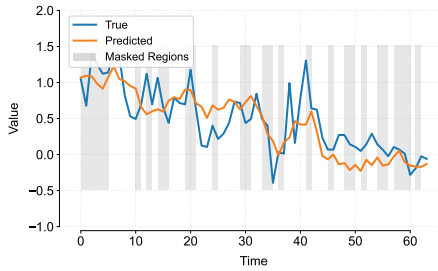
2) *Initialization of prompt vector*: In RWKV-TS+, the prompt embedding X^{pt} primarily serves to integrate historical information and requires no parameter updates. However, an initial assignment must be performed when defining this embedding. This study investigates the impact of different

TABLE V
COMPARISON RESULTS FOR **IMPUTATION TASK** WITH 10 BASELINES. THE $\{12.5\%, 25\%, 37.5\%, 50\%\}$ TIME POINTS OF 96-LENGTH SEQUENCES ARE RANDOMLY MASKED. THE MSE AND MAE METRICS RESULTS ARE AVERAGED FROM 4 DIFFERENT MASK RATIOS.

Methods		ETTh1		ETTh2		ETTm1		ETTm2		ECL		Weather		Average	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
MLPs	DLinear	0.201	0.306	0.142	0.259	0.093	0.206	0.096	0.208	0.132	0.260	0.052	0.110	0.119	0.224
	LightTS	0.284	0.373	0.119	0.250	0.104	0.218	0.046	0.151	0.131	0.262	0.055	0.117	0.123	0.228
CNNs	TimesNet	0.078	0.187	0.049	0.146	0.027	0.107	0.022	0.088	0.092	0.210	0.030	0.054	0.049	0.132
Transformers	PatchTST	0.115	0.224	0.065	0.163	0.047	0.140	0.029	0.102	0.072	0.183	0.034	0.055	0.060	0.144
	ETSformer	0.202	0.329	0.367	0.436	0.120	0.253	0.208	0.327	0.214	0.339	0.076	0.171	0.197	0.309
	FEDformer	0.117	0.246	0.163	0.279	0.062	0.177	0.101	0.215	0.130	0.259	0.099	0.203	0.112	0.229
	Stationary	0.094	0.201	0.053	0.152	0.036	0.126	0.026	0.099	0.100	0.218	0.032	0.059	0.056	0.142
	Autoformer	0.103	0.214	0.055	0.156	0.051	0.150	0.029	0.105	0.101	0.225	0.031	0.057	0.061	0.151
	Informer	0.161	0.279	0.337	0.452	0.071	0.188	0.156	0.292	0.222	0.328	0.045	0.104	0.165	0.273
RNNs	RWKV-TS	0.268	0.310	0.060	0.164	0.149	0.219	0.030	0.111	0.180	0.293	0.038	0.066	0.120	0.193
	RWKV-TS+	0.266	0.313	0.055	0.154	0.133	0.207	0.028	0.105	0.167	0.280	0.035	0.060	0.114	0.186
$\downarrow_{RWKV-TS}$		$\downarrow$ 10.7%												$\downarrow$ 5.0%	$\downarrow$ 3.6%



(a) PatchTST



(b) RWKV-TS+

Fig. 8. Visualization of the ETTm1 dataset for the PatchTST and RWKV-TS+ algorithms in the **Time Series Imputation** task. The blue lines represent the ground truth sequence and the orange represents the predicted results. The gray interval indicates the masked area.

initialization methods on downstream task performance, as following the Tab. VI and VII. Four common initialization strategies *Zeros*, *Normal*, *Rand*, *Ones* are compared across six datasets spanning three distinct task categories.

From a task-specific perspective, long-range time-series forecasting and sequence classification tasks consistently demonstrated optimal performance when prompt embeddings were initialized to all zeros (*Zeros*). For instance, on the

ETTh1 dataset, this approach achieved a minimum MAE of 0.4442. In contrast, anomaly detection tasks exhibited no single optimal initialization strategy. On the SMAP dataset, both *Zeros* and *Ones* initialization yielded the highest F1-score. Conversely, performance on the SWaT dataset was superior under *Normal* and *Rand* initialization. These results suggest that introducing appropriate noise during initialization can be beneficial for enhancing performance in anomaly detection tasks.

TABLE VI
PERFORMANCE COMPARISON USING DIFFERENT INITIALIZED WAYS IN **LONG-TERM FORECASTING TASK**.

Initialization Way		<i>Zeros</i>	<i>Normal</i>	<i>Rand</i>	<i>Ones</i>
ETTh1 (L)	MSE	0.4440	0.4454	0.4455	0.4465
		(0.0028)	(0.0014)	(0.0014)	(0.0021)
	MAE	0.4464	0.4498	0.4473	0.4478
		(0.0022)	(0.0006)	(0.0006)	(0.0013)
ETTh2 (L)	MSE	0.3521	0.3567	0.3568	0.3567
		(0.0010)	(0.0011)	(0.0017)	(0.0023)
	MAE	0.3928	0.3948	0.3949	0.3956
		(0.0008)	(0.0003)	(0.0008)	(0.0016)

F. Comparison of Efficiency Analysis

RWKV-TS+ exhibits time and space complexity equivalent to that of RWKV-TS, namely $\mathcal{O}(L)$ for time complexity and $\mathcal{O}(L)$ for memory complexity. This is fundamentally attributed to its linear RNN architecture, which is superior to the $\mathcal{O}(L^2)$

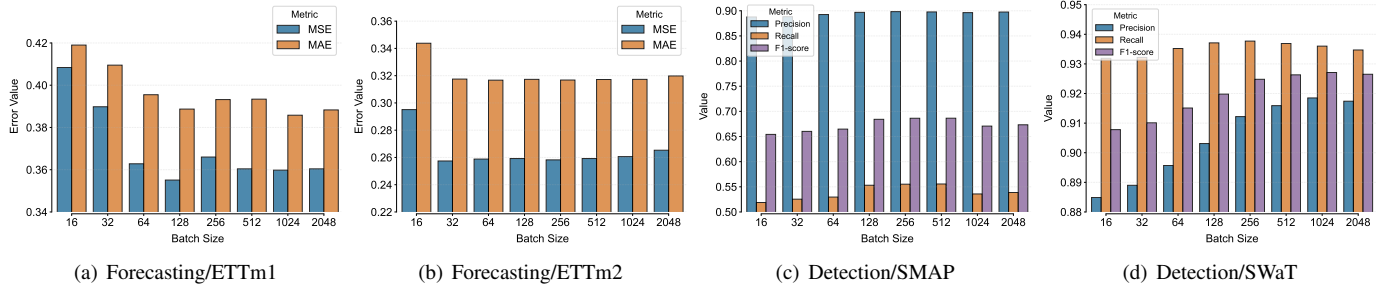


Fig. 9. Performance comparison with different batch size in **Long-term Forecasting** and **Anomaly Detection** tasks.

TABLE VII

PERFORMANCE COMPARISON USING DIFFERENT INITIALIZED WAYS IN **CLASSIFICATION** AND **ANOMALY DETECTION** TASKS.

Initialization Way		<i>Zeros</i>	<i>Normal</i>	<i>Rand</i>	<i>Ones</i>
PEMS-SF (C)	ACC	0.8439	0.8150	0.8150	0.7919
SR-SCP1 (C)	ACC	0.9215	0.9146	0.9147	0.9180
SMAP (A)	F1	0.8317	0.8228	0.8228	0.8317
SWaT (A)	F1	0.9718	0.9722	0.9722	0.9717

TABLE VIII

COMPARISON OF PARAMETER COUNT, INFERENCE COSTS OF TEST SET, AND MEMORY: THE INFERENCE TIME IS MEASURED BY THE WHOLE TEST SET.

Models	Parameters	Inference Time(s)	Memory
RWKV-TS+(best)	0.794M	5.646	1156M
RWKV-TS+	0.794M	4.095	1280M
RWKV-TS	0.794M	3.155	1184M
FEDformer	2.901M	10.807	7111M
PatchTST	1.179M	3.305	1952M

complexity of the vanilla Transformer and comparable to the highly efficient DLinear model. Moreover, since no additional parameters are introduced, RWKV-TS+ maintains the same parameter count as RWKV-TS.

Furthermore, using the ETTh2 dataset (designed for long-term forecasting) as an example, we set the batch size to 256, the hidden dimension to 128, and employed 3 encoding layers. The model was benchmarked against FEDformer and PathTST in terms of parameter count, inference time, and training memory consumption, with results summarized in Table VIII. Evidently, both RWKV-TS+ and RWKV-TS possess identical parameter counts, which are lower than those of other baselines. GPU memory consumption during training is also lower compared to other baselines, albeit slightly higher than that of RWKV-TS—primarily due to the caching of batch-level prompt vector matrices. Additionally, inference time on the test set is longer than that of RWKV-TS, owing to the pre-execution of intra-batch information gathering and inter-batch information integration operations, which duplicate the time-mixing operation twice. We also report the efficiency of the proposed model under optimal performance conditions (with a batch size of 64). In summary, RWKV-TS+ demonstrates highly competitive efficiency compared to other baselines, making it well-suited for deployment on edge devices.

G. Comparison for Different Prompt Modes

To evaluate the efficacy of integrating historical information through dynamic prompts in RWKV-TS+, we conduct a comparative analysis with several representative algorithms on long-term forecasting tasks, as summarized in Table IX. LLM-Time [48] employs manually designed static prompts, TEMPO [38] utilizes self-learned static prompts, while TSF-Prompt [40] incorporates a hybrid of multiple static prompting

TABLE IX

COMPARISON RESULTS BETWEEN W/ AND W/O PROMPT-BASED METHODS IN **LONG-TERM FORECASTING** TASKS. ALL THE RESULTS ARE AVERAGED FROM 4 DIFFERENT PREDICTION LENGTHS, THAT IS {96, 192, 336, 720}. THE BEST MSE AND MAE ARE BOLD AND RED AND THE 2ND BEST ONES VIA BOLD AND BLUE.

Methods		w/ Prompt				w/o Prompt	
		RWKV-TS+	TSF-Prompt	Time-LLM	TEMPO	GPT2	T5
ETTh1	MSE	0.444	0.433	0.449	0.428	0.436	0.428
	MAE	0.446	0.438	0.445	0.427	0.436	0.433
ETTh2	MSE	0.352	0.369	0.380	0.361	0.382	0.396
	MAE	0.392	0.398	0.403	0.398	0.407	0.411
ETTm1	MSE	0.355	0.381	0.421	0.501	0.555	0.580
	MAE	0.388	0.399	0.413	0.458	0.485	0.498
ETTm2	MSE	0.257	0.283	0.291	0.280	0.289	0.287
	MAE	0.317	0.329	0.335	0.328	0.333	0.329
Weather	MSE	0.249	0.254	0.266	0.282	0.356	0.338
	MAE	0.290	0.279	0.285	0.316	0.347	0.343
Traffic	MSE	0.392	-	0.440	0.503	0.554	0.540
	MAE	0.267	-	0.301	0.358	0.397	0.384
ECL	MSE	0.160	-	0.168	0.216	0.223	0.221
	MAE	0.252	-	0.270	0.308	0.315	0.314

strategies. All comparative results are extracted from their respective original studies, with the performance of GPT2 [49] and T5 [50] drawn from [38]. RWKV-TS+ achieves the SOAT performance across multiple datasets, demonstrating its robustness and stability. These suggest that dynamic prompts effectively capture feature dependencies within local temporal windows, thereby better facilitating long-term forecasting capabilities of the model.

VI. CONCLUSION

In response to the challenge of insufficient temporal information integration in current models, this work identifies that the issue primarily stems from the batched training paradigm. Inspired by prompt-based learning, we propose the RWKV-TS+ model, building upon the baseline RWKV-TS framework. RWKV-TS+ incorporates an adaptive long-range prompting mechanism that enables inter-batch temporal feature aggregation. Specifically, the Patch Features Collector to gather and compress patch-level features within the same batch, and a Time Features Aggregator to perform cross-batch blending for integrating inter-batch prompts over extended temporal scales. The Long Prompt Operator then incorporates the fused prompts into corresponding batches, effectively guiding each batch's processing.

To comprehensively evaluate the efficacy of the proposed method, we conduct extensive experiments on 34 benchmarks across four downstream temporal tasks (Long-term Forecasting, Classification, Anomaly Detection, and Imputation), and compare against the ten SOTA algorithms. Experimental results demonstrate the competitive performance of RWKV-TS+ and notable improvements over the original RWKV-TS. Furthermore, we provide theoretical analysis on the feasibility and consistency of integrating longer-range temporal information via long-range prompts. Importantly, RWKV-TS+ enhances representational capacity without introducing any additional parameters, ensuring deployment-ready model conditions. Compared to static prompts, the proposed dynamic prompts embedded with historical information more effectively enhance model reasoning.

However, this study does not explore the impact of integrating multiple types of prompts on temporal representation learning. Future work may investigate adaptively providing appropriate prompts from multiple modalities for different batches as a promising research direction.

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